

From Working Prototype to Submitted Paper

A senior advisor roadmap for a KAN-based decision support system for HiPCO SWCNT reactor quality prediction

Advisory review prepared for the NoPo Nanotechnologies reactor-AI project • Scope: process systems engineering + machine learning publication • Target: IEEE-style conference paper, 6 to 10 week horizon

This document is written to be useful before you write a single sentence of the paper. The most valuable thing I can do as an advisor is tell you plainly where your novelty is real and where it is thin, because the reviewers will find the thin parts whether or not you do. The short version: no single component of your planned system is new anymore, but the specific combination and the domain are defensible if you frame them correctly. The rest of this guide shows you how.

READ THIS FIRST: THE HONEST HEADLINE

As of mid 2026, three things you might have hoped were novel are already published: (1) XGBoost predicting the G/D ratio for carbon nanotube synthesis, (2) KAN used as an interpretable industrial product-quality model, and (3) KAN forward models with gradient-based inverse design for materials. If you claim any of these as your core contribution, you will likely be desk-flagged or rejected. Your genuine, defensible contribution is narrower and is described in Section 1.

1. Novelty positioning

1.1 What already exists (literature check)

I grouped the prior art into the four buckets you asked about. The entries below are the ones a reviewer in this area will already know, so you must cite them.

(A) MACHINE LEARNING FOR CNT / SWCNT SYNTHESIS

- **Aerosol CVD, Nasibulin group (Aalto / Skoltech).** A widely cited line of work predicting SWCNT yield and quality from synthesis parameters using artificial neural networks on a few hundred data points (Iakovlev et al., *Carbon* 2019; the 369-point aerosol-CVD study). This establishes that small-sample ML for SWCNT process prediction is already a known result, including the "a few hundred points is enough" finding.
- **ACS Nano forest-synthesis study.** "Addressing the Trade-Off between Crystallinity and Yield in SWCNT Forest Synthesis Using Machine Learning" uses *XGBoost to predict the G/D ratio* and even runs virtual (inverse) experimentation to recommend conditions. This is the closest single paper to your forward model and you must position against it directly.
- **High-throughput + random forest (Nano Research, 2021)** mapping growth parameters to the G/D ratio. Reinforces that G/D prediction from process parameters is a solved problem class.

(B) INDUSTRIAL AI DECISION SUPPORT AND SOFT SENSORS

- **Fricz, Horvath, Kummer, "Kolmogorov-Arnold and deep learning networks for industrial explainable product quality prediction," *Digital Chemical Engineering* 18:100289 (2026).** This is the paper that most threatens a generic framing. It applies KAN as an interpretable soft sensor for product quality on a real industrial plant, with symbolic-regression interpretability. If your abstract says "we use KAN for interpretable industrial quality prediction," this paper already did it.
- **Guo et al., "Kolmogorov-Arnold network driven soft sensors for chemical processes," *Computers and Chemical Engineering* 210:109612 (2026)** and a growing set of KAN soft-sensor papers (temporal-convolutional KAN in IEEE Xplore; KAN autoencoders for data-scarce fault detection, McMaster / *Processes* 2025). The message: KAN for process data is now an active, crowded subfield, not an untouched idea.

(C) KAN APPLICATIONS AND (D) PHYSICS-INFORMED SURROGATE MODELING

- **Liu et al., "KAN: Kolmogorov-Arnold Networks," arXiv:2404.19756 (2024); "KAN 2.0," arXiv:2408.10205 (2024).** The originals. Interpretability via learnable splines and symbolic snapping are their claims, not yours.
- **Wang et al., "Kolmogorov-Arnold-Informed Neural Network (KINN)," *Computer Methods in Applied Mechanics and Engineering* 433:117518 (2025).** Physics-informed KAN solving forward *and inverse* problems. Directly relevant to your "differentiable inverse" pitch.
- **Fronzi et al., "KAN in Thermoelectric Materials Design," arXiv:2510.02681 (2025) and Tung et al., "Scattering-based structural inversion of soft materials via KAN," *J. Chem. Phys.* 162:074106 (2025).** Both do KAN forward prediction plus inverse / reverse-engineering of materials with interpretability. Your inverse-through-KAN idea is a known pattern in materials by now.

ALREADY PUBLISHED, SO CITE IT AS RELATED WORK, DO NOT CLAIM IT

XGBoost predicting G/D ratio (ACS Nano). KAN as an interpretable industrial quality soft sensor (Fricz 2026, Guo 2026). KAN forward-plus-gradient-inverse for materials design (KINN 2025, Fronzi 2025, Tung 2025). Small-sample ANN for SWCNT synthesis (Nasibulin group). Treat all five as related work you extend, not novelty you own.

1.2 The specific gap you actually fill

Strip away everything that is covered above and what remains is genuinely yours:

- **Process specificity.** Essentially all CNT-ML prior art is aerosol / floating-catalyst CVD or substrate-grown forests. **HiPCO (high-pressure carbon monoxide disproportionation) is a distinct, industrially important, and under-studied process** with its own parameter space (CO pressure, catalyst injector geometry, radial and axial metal gradients, Fe/Ni/Cr ppm). There is no published ML quality model for a HiPCO reactor. That is a real, if modest, domain gap.
- **Physics-engine-as-simulator under extreme data scarcity.** You have a validated 167-formula first-principles calculation engine and only 20 to 50 real batches. Using that engine as a *ground-truth synthetic-data generator* to bootstrap a differentiable surrogate, then validating on withheld real batches, is a defensible methodological contribution. Most CNT-ML papers simply have more data or accept overfitting risk. You are turning a proprietary-data constraint into a physics-grounded data-augmentation strategy.
- **End-to-end deployed DSS, not a benchmark.** OPC-UA historian ingestion, alarm and production-state reconstruction, header mapping of operator logs, the physics engine, the KAN forward model, and a differentiable inverse recipe recommender, all wired into an operator-facing tool. This is a *systems and application* contribution: an industrial digital-advisor case study end to end. Reviewers at application venues value this even when each component is known.
- **Closed-loop inverse recommendation with physics feasibility.** Doing the inverse optimization *through* the same differentiable KAN, constrained to the physically feasible region defined by your engine, so the output is an actionable setpoint recipe rather than an abstract material target.

NOVELTY STATEMENT (USE THIS AS THE CORE CLAIM OF THE PAPER)

We present the first end-to-end machine-learning decision support system for a high-pressure carbon monoxide (HiPCO) single-walled carbon nanotube reactor, spanning OPC-UA historian ingestion, first-principles feature computation, forward quality prediction, and inverse recipe recommendation. To overcome extreme batch-level data scarcity, we use a validated 167-formula process engine as a physically grounded synthetic-data generator and train a Kolmogorov-Arnold Network (KAN) surrogate whose learnable spline activations expose process-to-quality relationships and whose differentiability enables gradient-based inverse optimization of setpoints under physics-feasibility constraints. We benchmark the KAN against the deployed XGBoost and PLS baselines on withheld real batches, and report the interpretability structure recovered from the reactor. The contribution is not a new learning algorithm but a validated, deployable integration that makes differentiable, interpretable quality control tractable for a proprietary, low-data industrial nanomaterials process.

Note the deliberate hedging: "first end-to-end DSS for a HiPCO reactor" and "not a new learning algorithm but a validated deployable integration." Both are true and both are hard to attack. Do not upgrade these claims to "novel KAN architecture" or "novel inverse method." You will lose that fight.

2. Technical roadmap (current system to KAN architecture)

2.1 Forward prediction as a KAN

Keep your existing feature vector: process setpoints plus the 11 physics-engine secondary parameters (flow velocity, heat loss, catalyst concentration, residence time, and the rest). Targets are your quality metrics (G/D ratio, UV-Vis purity, Fe/Ni/Cr ppm axial and radial). Concrete steps:

1. Use the `pykan` reference implementation (or an efficient variant such as an RBF-KAN / FastKAN, or Fourier-KAN) so training stays fast on your small data. Start with a shallow network, one or two KAN layers, grid size 3 to 5. Deep KANs overfit tiny datasets.
2. Wrap inputs and outputs in a `StandardScaler` pipeline exactly as you already do, so the KAN and the baselines see identical preprocessing. This keeps the comparison honest.
3. Train with the same cross-validation strategy you already adapted for low sample count. Report CV R2 and NMAE per target, matching your existing metric set, so numbers are comparable across models.
4. Apply KAN pruning and symbolic snapping (KAN 2.0 style) after training to extract candidate closed-form activations. These become your interpretability figures.

2.2 Gradient-based inverse optimization through the KAN

This is where the KAN earns its place over XGBoost. Because the forward KAN $f(x)$ is differentiable end to end, you can solve for setpoints that hit a desired quality target by gradient descent on the input:

- Fix the trained forward KAN. Define a loss $L(x) = || f(x) - y_{\text{target}} ||^2 + \text{lambda} * \text{penalty}(x)$, where `penalty(x)` enforces box bounds and physics feasibility (see 2.4).
- Optimize x with Adam or L-BFGS from several physically plausible initial points, keeping the best feasible solution. Multi-start avoids local minima and gives you an uncertainty spread on the recommendation.
- This directly replaces the PLS analytical inverse and the KNN-in-output-space backtracker with a single principled optimizer that covers all input parameters at once, which is exactly the direction your own solver iterations were already heading.

2.3 Whether and how to ensemble with XGBoost and PLS

Do not drop the baselines. Reframe them as the incumbents you must beat and as an ensemble option:

- **Primary story: KAN as the deployable single model**, with XGBoost and PLS as baselines in your comparison table. This is the cleanest narrative.
- **Optional robustness result:** a simple average or stacked ensemble of KAN and XGBoost for the forward task. On tiny data, an ensemble often reduces variance and gives you a defensible "best forward model" number. Report it as an ablation, not the headline.
- **Keep PLS only as the inverse baseline** so you can show the differentiable KAN inverse beating the linear analytical inverse on recipe error. That contrast is a good figure.

2.4 Integrating the 167-formula physics engine

You have three options. Pick based on what is defensible in the time you have.

Approach	What it means	Advisor verdict
A. Physics as features	Engine outputs (residence time, catalyst concentration, etc.) are appended to the KAN input vector. You already do this.	Lowest risk, defensible, do this as the baseline. Weakest novelty on its own.
B. Physics as synthetic-data generator	Sample setpoints, run the engine, produce physically consistent (x, secondary-params) pairs to pre-train or augment the KAN, then fine-tune on real batches.	Recommended headline. This is the most defensible and directly answers "why so little data." Ties Section 2 to Section 3.
C. Physics as a soft constraint / loss term	Add a penalty when KAN predictions violate engine-derived relationships (a physics-informed KAN, in the KINN spirit).	Highest novelty but highest risk and time cost. Attempt only if A and B are done and you have a spare week. Frame as future work otherwise.

For a conference paper on your timeline, **lead with B, keep A as the always-on baseline, and mention C as an ablation or future work.** Claiming a full physics-informed KAN (C) invites comparison to KINN and mechanics-PINN literature you do not have time to match.

2.5 Experiments needed for a complete results section

Experiment	What it produces	Priority
Forward model comparison	Table: KAN vs XGBoost vs PLS (and MLP) on CV R2 and NMAE per quality target, on synthetic and on real held-out batches.	Essential
Inverse recipe comparison	Table: KAN gradient inverse vs PLS analytical inverse vs KNN backtracker, measured by recipe error and by re-simulated quality error through the engine.	Essential
Data-scarcity ablation	Curve: accuracy vs number of real batches, with and without synthetic pre-training. Shows the physics-augmentation value.	Essential
Interpretability	Plots of learned spline activations for the most influential inputs, plus any symbolic forms recovered. Cross-check against known HiPCO physics.	Essential
Noise / robustness ablation	Performance under injected sensor noise and missing-data patterns (ties to Section 3).	Strong to have
Ensemble ablation	KAN alone vs KAN+XGBoost stack.	Optional

3. Synthetic data strategy

3.1 Build the dataset from your engine as ground truth

1. **Define the input sampling space.** For each real setpoint (CO pressure, catalyst flow, injector geometry, water flow, temperatures) set a plausible range and sample with Latin Hypercube or Sobol sampling to cover the space evenly. Do not sample uniformly and independently if the real variables are correlated (see 3.3).
2. **Run the 167-formula engine on each sampled point** to compute the 11 secondary parameters. This gives you physically consistent (setpoint, derived-parameter) records at any volume you want.
3. **Generate quality targets** either from a physically motivated response surface you fit to your real data (kept private) or by training an intermediate model on real data and using it only to label synthetic points. Disclose which you used. The cleanest, most defensible route is a documented parametric response model with literature-bounded coefficients.
4. **Never publish real values.** Report only ranges, distributions, and normalized quantities. The synthetic set is what appears in the paper; the real set stays behind the proprietary wall and is used only for the held-out validation numbers.

3.2 Literature values to bound the ranges (so reviewers accept realism)

Anchor your synthetic ranges to published HiPCO and SWCNT numbers so a reviewer cannot say "these are made up." Useful anchors: original HiPCO process reports (Nikolaev, Smalley and co-workers) for CO pressure and temperature regimes; SWCNT G/D ratios typically reported from roughly 2 up to the tens for high-quality material; Raman-based crystallinity conventions; and the aerosol-CVD ML datasets (Nasibulin group) for the shape of parameter-to-quality relationships. Cite these when you state each range. The point is not that your synthetic values equal theirs, it is that they fall inside physically published envelopes.

3.3 Make the data not "too easy"

A reviewer's fastest rejection is "the synthetic data is trivially learnable." Prevent it:

- **Correlated inputs.** Impose a realistic covariance structure (for example temperature and pressure co-vary, catalyst flow correlates with injector geometry) rather than independent draws. Sample from a multivariate distribution with an estimated correlation matrix (report the matrix shape, not the raw data).
- **Heteroscedastic noise.** Add measurement noise whose magnitude depends on the operating regime, and add label noise on the quality targets sized to real lab-measurement repeatability (G/D ratio and ppm values have non-trivial measurement error). Justify the noise levels from instrument specs.
- **Missing-data patterns.** Mirror the real gaps: empty Data Logger fields, sensor dropouts, batches missing UV-Vis or ICP results. Inject the same missingness structure so the model is tested on realistic incompleteness.
- **Regime imbalance.** Real production clusters around a few recipes. Make the synthetic distribution non-uniform so the tails are sparse, matching reality and making the task honestly hard.

3.4 How to structure the paper's data section

Split it explicitly into two subsections. **Synthetic methodology (fully disclosed):** sampling scheme, engine role, range table with literature citations, noise and missingness models, correlation structure. This is what makes the work reproducible in principle. **Real-data validation (results only, no values):** state sample count as a range (for example "20 to 50 production batches"), describe the withheld-batch protocol, and report only aggregate metrics (R2, NMAE). Add one sentence stating the real data is proprietary and cannot be released, and that all releasable artifacts are synthetic. Reviewers accept this when the synthetic methodology is thorough enough to stand alone.

4. Validation plan

4.1 Minimum viable validation that meaningfully strengthens the paper

The single most valuable experiment is **one closed-loop confirmation run**: take the inverse optimizer's recommended setpoints for a target quality, run one real HiPCO batch at those setpoints, and compare measured quality to the predicted target. Even a single successful closed-loop point moves the paper from "we predicted on held-out data" to "the recommendation worked on the real reactor," which is a qualitatively stronger claim and is often the difference between a poster and an oral.

4.2 If a real run is not feasible before the deadline

This is common and not fatal. Do the following instead and frame it honestly:

- **Simulated closed loop.** Feed the inverse recommendation back through the physics engine and report how close the engine-computed quality lands to the target. State clearly that this is engine-in-the-loop validation, not reactor-in-the-loop.
- **Held-out real-batch validation** for the forward model, reported as aggregate metrics only.
- **An explicit limitations paragraph** (see Section 7) that names the absence of a physical closed-loop run as the primary limitation and states it as planned future work. Reviewers reject papers that hide missing validation; they usually accept papers that name it precisely and show strong offline evidence plus a credible plan.

5. Paper structure (IEEE conference format)

Target 6 pages of content plus up to 2 pages allowed by most IEEE conference templates (double column). Word counts below assume the two-column IEEE style.

Section	Length	Contents
Abstract	~180 words	Problem, the DSS, the physics-augmented KAN idea, headline results, one-line significance.
I. Introduction	~0.75 col	HiPCO and why quality control is hard; data scarcity; the decision-support framing; contributions as a bullet list of 3 to 4 items.
II. Related Work	~0.75 col	The four buckets from Section 1.1. Be generous with citations here; it is your defense.
III. System and Data Pipeline	~1 col	OPC-UA ingestion, preprocessing, header mapping, the 167-formula engine. One architecture figure.
IV. Synthetic Data Methodology	~1 col	Section 3 content: sampling, ranges with citations, noise, missingness, correlation.
V. KAN Forward and Inverse Models	~1.25 col	Forward KAN formulation; gradient inverse with feasibility constraints; ensemble note.
VI. Experiments and Results	~1.75 col	The six experiments from Section 2.5. Tables and figures carry this section.
VII. Interpretability	~0.5 col	Spline-activation plots and any symbolic forms, cross-checked with physics.
VIII. Limitations and Future Work	~0.4 col	Honest limitations (Section 7), physical closed-loop run as future work.
IX. Conclusion	~0.25 col	Restate the contribution as an integration and deployment result.

5.1 Essential figures

1. **System architecture diagram** (historian to physics engine to KAN forward and inverse to operator). This is the figure that sells the systems contribution.
2. **KAN learned-activation plots** for the top few inputs, the visual proof of interpretability.
3. **Forward-vs-inverse performance figure:** predicted vs actual for the forward model, and recipe-error comparison for the inverse.
4. **Data-scarcity ablation curve** (accuracy vs number of real batches, with and without synthetic pre-training).
5. **Uncertainty bands** on the inverse recommendation from multi-start optimization.

5.2 Suggested title and abstract

Title: "A Physics-Augmented Kolmogorov-Arnold Decision Support System for Quality Prediction and Inverse Recipe Recommendation in a HiPCO Carbon Nanotube Reactor."

DRAFT ABSTRACT (ABOUT 180 WORDS)

High-pressure carbon monoxide (HiPCO) synthesis produces high-quality single-walled carbon nanotubes, but batch-to-batch quality is difficult to predict and control, and production data is scarce and proprietary. We present an end-to-end decision support system that ingests OPC-UA reactor historian data, reconstructs production state, computes secondary process parameters through a validated 167-formula first-principles engine, and predicts nanotube quality (G/D ratio, optical purity, and metal impurity levels). To address extreme data scarcity, we use the process engine as a physically grounded synthetic-data generator, bounding all ranges by published HiPCO and SWCNT values and injecting realistic noise, correlation, and missingness. On this data we train a Kolmogorov-Arnold Network whose learnable spline activations expose interpretable process-to-quality relationships and whose differentiability enables gradient-based inverse optimization of setpoints under physics-feasibility constraints. Against deployed XGBoost and PLS baselines evaluated on withheld real batches, the system matches or improves forward accuracy and yields lower inverse recipe error. The result is a deployable, interpretable, and differentiable quality-control tool for a low-data industrial nanomaterials process.

6. Venue recommendation

TIMING REALITY CHECK

It is mid August 2026. Most strong IEEE applied-ML conferences for the 2026 cycle have already closed (ICMLA 2026 closed around mid May; IEEE CAI 2026 and several IEEE industrial-AI conferences are past). For a submission 6 to 10 weeks out you are realistically choosing among workshop tracks, rolling or extended Scopus/IEEE conferences, and fast-review journals. If paper quality matters more than speed, aim for a first-tier applied venue in the next cycle instead.

Venue	Type / selectivity	Fit	Timing note
NeurIPS / ICML "ML and the Physical Sciences" or "AI for Materials" workshops	Workshop, moderate, non-archival or lightly archival	Excellent fit for early, honest, systems-plus-method work. Values novel applications over SOTA-chasing. Ideal for student-level KAN-for-materials.	Fall 2026 workshop deadlines are the realistic near-term target. Check the specific workshop CFP.
IEEE ICMLA (Int'l Conf. on Machine Learning and Applications)	IEEE, applications-focused, mid selectivity	Strong fit: explicitly welcomes ML applied to industry and manufacturing; IEEE Xplore indexed; accepts short (poster) papers, good for students.	2026 cycle closed (~May). Realistic target is ICMLA 2027 if you can wait; otherwise use as the model for a similar open venue.
Rolling / extended Scopus-IEEE applied-AI conferences (AIITA, AAIEE, CAIS and similar)	IEEE/Scopus, lower selectivity, fast turnaround	Reasonable if you need a proceedings paper this cycle. Quality varies; verify indexing and that it is genuinely peer reviewed before paying fees.	Several run multiple cycles with late/extended deadlines in fall 2026. Confirm each on its official CFP.
Fast-review journals (e.g. MDPI <i>Processes</i> special issues, Elsevier <i>Digital Chemical Engineering</i>)	Journal, rolling, rapid	Good alternative if no conference deadline fits. <i>Digital Chemical Engineering</i> already published closely related KAN soft-sensor work, so it is a receptive, on-topic home.	Rolling submission; no fixed deadline pressure.
AIChE Annual Meeting (session on data-driven / ML process systems)	Domain conference, abstract-based	Excellent domain audience for a process-systems framing. Presentation-first (proceedings paper optional).	Abstract deadlines are typically spring for the November meeting, so plan for the next cycle.

Advisor recommendation: for the 6 to 10 week horizon, submit to a fall 2026 physical-sciences / materials ML workshop (best signal-to-effort for a student contribution), and in parallel prepare the fuller version for ICMLA 2027 or a *Digital Chemical Engineering* submission. Always verify a venue's peer-review integrity and indexing on its official site before

committing, especially for any conference soliciting fees with unusually late deadlines.

7. Risk list: the reviewer objections to preempt

Likely objection	How to preempt it in the writing
"The synthetic data makes the results circular. You train and test on your own engine."	Report the key metrics on <i>withheld real batches</i> , not synthetic. State explicitly that synthetic data is for training and augmentation, and that all headline accuracy numbers come from real held-out data. Include the data-scarcity ablation to show synthetic pre-training helps on real data.
"KAN for interpretable industrial quality prediction is already published."	Cite Frizc 2026 and Guo 2026 in Related Work and state your delta clearly: a new process (HiPCO), the physics-engine-as-generator strategy, and the deployed inverse-recommendation loop. Do not claim the KAN method itself as novel.
"No physical validation. You never ran the recommended recipe on the reactor."	If you cannot run one closed-loop batch, report engine-in-the-loop validation, name the missing reactor run as the primary limitation in Section VIII, and present it as concrete, scoped future work. Honesty here is protective; concealment is fatal.
"Sample size of 20 to 50 batches is too small for any credible ML claim."	Frame small-sample as the central research problem, not a weakness to apologize for. Cite the Nasibulin-group finding that a few hundred (and sometimes fewer) points can suffice for SWCNT process ML, use rigorous cross-validation, report variance not just point estimates, and let the physics-augmentation be the answer to scarcity.

Week-by-week plan (about 10 to 15 hours per week)

This is a 9-week plan ending in a submitted paper. A compressed 6-week variant is noted at the end.

Week	Focus	Concrete deliverable
1	Lock the novelty framing and finish the literature map. Read the five threatening papers in Section 1 in full. Freeze the novelty statement.	Related-work notes with citations; one-paragraph novelty statement finalized.
2	Build the synthetic-data generator: LHS/Sobol sampling, engine runs, correlation and noise and missingness models, literature-bounded ranges.	Reproducible synthetic dataset plus the range table with citations.
3	Implement the forward KAN (pykan or FastKAN). Match your existing preprocessing and CV. Get first CV R2 and NMAE numbers.	Working forward KAN; first comparison numbers vs XGBoost and PLS.
4	Implement the gradient-based inverse through the KAN with feasibility constraints and multi-start. Compare against PLS and KNN inverse.	Inverse optimizer; inverse-comparison table (recipe error + re-simulated quality error).
5	Run the ablations: data-scarcity curve (with/without synthetic pre-training), noise-robustness, optional ensemble.	All ablation results and plots.
6	Interpretability: pruning, symbolic snapping, spline-activation plots. Cross-check against known HiPCO physics. Real held-out validation numbers.	Interpretability figures; final real-batch validation metrics.
7	Make the five essential figures publication-clean. Draft Sections III to VII (system, data, models, results, interpretability).	All figures final; results half of the paper drafted.
8	Draft Introduction, Related Work, Limitations, Conclusion, and the abstract. Full internal read-through.	Complete first full draft.
9	Revise for the reviewer objections in Section 7, tighten to the page limit, format to the IEEE template, proofread, submit.	Submitted paper.

Compressed 6-week variant: merge weeks 1 and 2 (framing plus data generation), merge 3 and 4 (forward and inverse together), keep week 5 ablations but drop the ensemble, compress interpretability and validation into one week, then two weeks of writing and revision. Cut the noise-robustness ablation first if time runs short, and target a workshop rather than a full conference paper.

Advisor's closing note: your risk is not that the work is weak, it is that the obvious framing ("we applied KAN to nanotube data") is already taken. Win by being specific and honest: a real proprietary HiPCO process, a physics engine turned into a data generator to beat scarcity, and a deployed differentiable inverse recommender. Name your limitations before a reviewer does. That combination is publishable at the venues in Section 6, and the week-by-week plan above gets you there. Verify all conference deadlines and indexing on official sites before committing, as smaller venues change dates frequently.